Co1 assessment:constraint based problem solving

1)Hospital shift scheduling:

```
doctors = ["D1", "D2", "D3"]
shifts = ["Morning", "Afternoon", "Night"]
schedule = {}

def is_valid(doctor, shift):
    if shift in schedule.values():
        return False

    if doctor == "D1" and shift == "Night":
        return False

    if doctor == "D3" and shift == "Morning":
        return False

    temp = schedule.copy()
    temp[doctor] = shift

    if "D2" in temp and "D3" in temp:
        if shifts.index(temp["D2"]) >= shifts.index(temp["D3"]):
            return False

    return True

def backtrack(index):
    if index == len(doctors):
        return True

    doctor = doctors[index]
```

```
    for shift in shifts:
        print("Trying", doctor, "=", shift)

        if is_valid(doctor, shift):
            schedule[doctor] = shift
            print("Accepted:", schedule)

            if backtrack(index + 1):
                return True

            del schedule[doctor]
            print("Backtracking...")

        else:
            print("Rejected:", doctor, shift)

    return False

if backtrack(0):
    print("\nFinal Valid Schedule:")
    for doctor in doctors:
        print(doctor, "->", schedule[doctor])
else:
    print("No valid schedule found")
```